CSE 290T Final Report: Peer Prediction Markets

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Abstract. We define a *peer prediction market* as a prediction market for subjective information when objective truth is intrinsically or practically unknowable. We show that the underlying contracts of any peer prediction market are ultimately grounded in the behavior of the market itself (or closely related markets), and consider such markets as global games. We review the successful, limited implementation of prediction markets in the trust-less setting of cryptocurrency blockchains with Augur and highlight some of its limitations for our proposed setting. Finally, we consider routes for adapting the "Bayesian Truth Serum" consensus mechanism for closing peer prediction markets in incentive-compatible ways.

Keywords: Peer prediction markets · Augur · Global Game · Bayesian Truth Serum

1 Introduction

1.1 Motivation and Objective

Prediction markets (also termed markets for event derivatives) have long been recognized as effective mechanisms for information elicitation and aggregation [3] and have recently spawned derivative notions in computer science [10]. Such markets attract attention not only as prediction tools, but as potentially powerful institutions for economic risk-management [1] or market-driven governance [4].

The purpose of prediction markets is to elicit and aggregate predictions, beliefs, or private information in the service of collective decision making. When individual incentives are best realized by honest participation, the mechanism by which they participate is said to be *incentive-compatible*. It remains an **open question** whether markets may be appropriate or viable for aggregating *subjective* data in an incentive-compatible way. This question may be asked in terms of the viability of a prediction market without ground truth or in the *peer prediction setting*.

In the peer prediction setting, the true outcome of a posed event or observation is unknown, cannot be verified, or is ill-posed for independent, objective assessment. Nonetheless, we may wish to elicit and aggregate subjective information from a population, even without the means of independent verification: For example, we might say that the subjective quality of art or the result of a hypothetical but cost-prohibitive experiment have no actionable ground truth for the purposes of elicitation, but we still wish to know the private beliefs of a population. The leading techniques for eliciting such information involves scoring and rewarding participants for reports which are most informative of the ultimate consensus: an adaptation of proper scoring rules when ground truth is known.

We believe that the adaptation of market contracts to the peer prediction setting is possible, and the effort to do so is the purpose of this work. We will evaluate other mechanisms including Global Games, Bayesian Truth Serum, and Augur as references to address the open problem of using prediction market without ground truth.

2 Background and Related Work

2.1 Prediction Markets

For motivational purposes, we will limit our discussion to a prediction market which considers some yet undetermined binary event $e \in \{0, 1\}$. In this market, a contract may be proposed by any participant which is redeemable upon the determination of e. For simplicity, we specify

$$\text{dollar value for which contract is redeemable } V = \begin{cases} 1 & (e = 1) \\ 0 & (e = 0) \end{cases}$$

The contract may be sold by the issuer for some price $0 \leq p^* \leq 1$. For any agent who believes with prior probability p that $e = 1$, the expected redeemable value of the contract is therefore

$$\begin{aligned}\mathbb{E}[V] &= p(1) + (1-p)(0) \\ &= p\end{aligned}$$

A risk-neutral market participant with static beliefs participating in a liquid market will purchase contracts when $p^* < p$ and sell contracts when $p < p*$. Independent of a participant's risk preferences however, market participation is voluntary, and participants will only buy or sell securities when they believe it is in their interest to do so.

When participants have static beliefs, the market is therefore incentive-compatible in reaching an equilibrium price.[1] We interpret this equilibrium market price as the market consensus or aggregated probability for event e.

2.2 Scoring Rules

When eliciting information from agents, a scoring rule assigns a *score* to the information provided by an agent, and an agent is presumed to seek to maximize his or her score without external incentives. Ideally, higher scores should represent higher utility for the scorer as well as well as the agent: In the information-elicitation setting, this incentive-compatibility occurs when an agent's score is maximized by providing truthful information.

Strictly proper scoring rules assign strictly maximal scores to truthful reports (i.e. an agent will expect to receive a strictly lower score when misreporting her beliefs about some variable or event). For example, considering an agent's report $a \in [0, 1]$ and some Boolean variable $y \in \{0, 1\}$ (with value known only to the scorer), the Brier scoring rule assigns score S as

$$S(a, y) = 2ay - a^2$$

In expectation, this score is maximized when a is equal to an agent's perceived probability p that $(y = 1)$.

$$\begin{aligned}\mathbb{E}[S(a, y)] &= 2ap - a^2 \\ \frac{\mathrm{d}}{\mathrm{d}a}\mathbb{E}[S(a, y)] = 0 &\implies a = p\end{aligned}$$

We note here that for any strictly proper scoring rule S,

$$S'(a, y) = \alpha S(a, y) + \beta \quad (\alpha > 0) \tag{1}$$

is also strictly proper given that α and β are independent of a.

Typically, scoring rules require the actual value of a variable or event to be known to the scorer. When this is not true, we enter the peer prediction setting.

2.3 Peer Prediction

The peer prediction setting is defined by a lack of access to objective ground truth. Nonetheless, clever mechanisms [8], [5] still provide a means to elicit incentive-compatible (truthful) reports from a population by assigning variable rewards to each participant depending on the aggregate results. Of these various mechanisms, we will focus on Bayesian Truth (BTS) as adequate background for our project.

Suppose we want to elicit statements from a population of n participants pertaining to an event or variable with m discrete possibilities. We represent the space Ω of these possibilities as a set of orthonormal vectors in $\mathbb{R}^m$.

$$\Omega = \mathbf{e}_1, \mathbf{e}_2, ..., \mathbf{e}_m.$$

Each of our participants has two questions to answer:

[1] When participants update their beliefs in response to the market and external incentives exist, the market is not necessarily incentive compatible, as dishonest trades may be abused to spread misinformation. In general, the absence of external incentives is a critical assumption for proof of incentive-compatibility for any mechanism.

1. Of the possible answers in Ω, which answer to this question will be selected by participants at a rate most under-predicted by the (geometric) average answer to question 2?
2. For each possibility in Ω, what proportion of participants will choose it to answer question 1?

Formally, question 1 elicits an answer from respondent r of the form:

$$\mathbf{x}^r \in \{\mathbf{e}^1, \mathbf{e}^2, ..., \mathbf{e}^m\}$$

while question 2 elicits a response on the probability simplex

$$\mathbf{y}^r \in \mathbb{R}^m : \sum_{i=1}^{m} y_i = 1, y_i \geq 0$$

We define

$$\bar{\mathbf{x}} = \lim_{n\to\infty} \frac{1}{n} \sum_{r=1}^{n} \mathbf{x}^r$$

as the true answer to question 2, constructed from the set of responses to question 1. We define the aggregate response to question 2 as $\bar{\mathbf{y}}$ according to the component-wise geometric mean of response

$$\bar{y}_i = \lim_{n\to\infty} \left(\prod_{r=1}^{n} y_i^r \right)^{(1/n)} \iff \log \bar{y}_i = \lim_{n\to\infty} \frac{1}{n} \left(\sum_{r=1}^{n} \log y_i^r \right)$$

Each respondent is scored by

$$\sum_k x_k^r \log \frac{\bar{x}_k}{\bar{y}_k} + \alpha \sum_k \bar{x}_k \log \frac{y_k^r}{\bar{x}}$$

where $\alpha > 0$ parameterizes a family of scoring functions we may choose from. The first term is the *information score* and the second is the *prediction score*.

This mechanism elicits truthful statements in a Bayesian Nash Equilibrium [8] and provides a means to extract consensus from a group of peers which is more likely to be correct than simply taking the majority opinion (e.g., BTS may recommend a minority opinion as being correct by relying on additional respondent information). This consensus is used to score and reward participants for the most informative reports.

Use of Bayesian Truth Serum requires a central authority to pay participants, and may not budget-balanced (depending on the choice of α). Furthermore, BTS relies on information which must be privileged to the central authority and not shared with participants during the elicitation phase. These two points stand in contrast to what may be assumed by an unfettered market. Nonetheless, we show in section 3.1 that we cannot simultaneously demand total public information and a distributed contract-closing mechanism. We comment more on extensions of BTS to implement peer prediction markets in 3.3.

2.4 Global Games

Global Game were first introduced by Carlsson and van Damm in 1993 [2]. Global Games are also known as incomplete information games. This means players may not know the characters of other players, the strategy space, or the payoff function for the game they are playing. Alternatively, each player may have a noisy, private observation for each of these game attributes, from which they derive estimates of their own best next action.

In two-person 2 x 2 normal form games, we assume each player $i \in \{1, 2\}$, has two actions, α and β. Suppose that G is the set of all games. Hence, for $g \in G$, the action of player i is sampled from a probability distribution. Based on this assumption, we can give the following description of global games:

- We first choose a game g from the set G. Players don't know which game was chosen, or may privately observe the choice with some noise.
- Players observe other players' actions, again with possible noise.
- After each player observes the actions of other players, they may select their own next move according to their private information and beliefs.
- Payoffs depend on both the game g and the actions of other players.

2.5 Augur

Augur and Gnosis are the two most well known decentralised prediction markets platforms. They were both created on 2015. They are both leveraging the wisdom of the crowd and decentralised networks to create prediction marketplaces. Both have their own ways to design incentives to ensure markets are correctly settled. In this report, however, We will only take a closer look at how Augur handles this.

Before diving into the details of Augur, some of the main concepts of Augur, such as peer prediction market and oracles, are briefly introduced to give the readers a general view of the Augur's architecture and the way it works.

Augur is a trustless, decentralized oracle and platform for prediction markets. In Augur, the participants who hold Augur's native Reputation token choose the outcomes of the Augur's prediction market. They do this through staking their tokens on the actual observed outcome and in return they get some settlement fees from the market. This strategy is designed to incentivized the reporters to act honestly since it is the most profitable option for Reputation token holders.

Token holders can post progressively-larger Reputation bonds to dispute proposed market outcomes. If the size of these bonds reaches a certain threshold, Reputation splits into multiple versions or forks, one for each possible outcome of the disputed market; In this case, token holders must choose one of them and exchange their Reputation tokens for that version. Finally, versions of Reputation which do not correspond to the real-world outcome will become worthless, as no one will participate in prediction markets unless they are confident that the markets will resolve correctly [7].

In this way, token holders will select the only version of Reputation which they know will continue to have value: the version that corresponds to reality. In this way, the Augur mechanism handles the possible attacks in the market.

Fig. 1. Simplified outline of the lifetime of a prediction market. [7]

Augur markets follow a four-stage progression: creation, trading, reporting, and settlement.

Anyone can create a market based on any real-world event. For this aim, they need Augur's native token, which is called Reputation (REP). These tokens are also needed by reporters when they report on the outcome of the markets created on the Augur platform. Reporters report on a market through staking their REP on one of the market's possible outcomes. In fact, they stake their REP on the outcome that they believe is going to match the real-world outcome of the market's underlying event. The market's final outcome is chosen by the majority role. In other words, the consensus of a market's reporters is considered the "truth" for the sake of determining the outcome of the market. Finally, the REPs of those reporters who have not staked on the consensus outcome is going to redistributed on the reporters who are on the consensus outcome side. In this way, Augur encourage the reporters to report on the truth.

Although REP plays a central role in Augur's operations, it is not used to trade in Augur's markets. Traders will never need to own or use REP, as they are not required to participate in the reporting process [7].

2.5.1 Oracles Smart contracts on permission-less blockchain, such as Etherium, are guaranteed to be tact through broadcasting one-way deterministic transactions, which should be verified by the majority of the participants. In this way, it is guaranteed that there is no Ctlr+Z in the blockchain world [9].

However, in most real-world cases, the smart contracts need to take inputs from the outer world, where it has no knowledge about. In this case we are faced with the element of trust, which the outcome of the smart contract depends on the external world of the blockchain.

The things become more complicated by knowing the fact that the miners on a permission-less blockchain cannot deterministically verify all external inputs, and will therefore allow execution of anything which matches the predefined criteria of the smart contract [9].

To make things more clear, let's take a very simple example: a smart contract allowing Alice and Bob to bet on the temperature of Santa Cruz one Bitcoin on 3 December 2019 at 6 PM. Alice and Bob each deposit 1 ETH in escrow in this smart contract; Alice receives 2 ETH if the temperature is above 50 °F/ 10 °C, else Bob receives the 2 ETH.

As it is mentioned before, the smart contract has no notion of the temperature of Santa Cruz, so we must find a way to ensure only the correct temperature is reported, especially since the outcome of the bet is irreversible once settled. For that purpose, we need oracles.

An oracle is an entity which signs claims about the state of the world. For instance, an oracle might report the 2019 European Champions' League winner or the winner of the US presidency election in 2020. The information may be retrieved or aggregated from one or multiple trusted sources by one or multiple oracles [9].

Before diving into the various technical challenges of trust-less oracles, it is worth mentioning some key use cases for oracle-powered smart contracts that we might encounter in the real-world applications [9].

- Financial products: smart contracts are a perfect match for many financial products such as interest rates swaps. Also, to settle the contract, most of them require some knowledge from the outer world to ensure appropriate on-chain execution. We already have some real-world platforms that make use of oracles for financial products such as CDx, dYdX, MakerDAO, Vega Protocol, and 0x protocol.
- Insurance Smart Contracts: The idea of replacing the conventional insurance with more reliable ones With trustless and reliable source of truth, which products could be bought and settled programmatically with a smart contract. The biggest concern here is fraud, and Oracles are the solutions here. Automatic payments if flight is delayed with Etherisc can be an example of Smart Contracts With trust-less and reliable source of truth.
- Shipment: GPS is not always suitable for a dApp requiring reliable location to automatically release a payment, because it is a centralised point of failure that can be spoofed. With FOAM for example, it is possible to lower this trust criteria and rely on a decentralised network, here acting as a location oracle, to guarantee a claimed location.
- Prediction markets: Decentralised prediction markets (Augur, Gnosis) have leveraged unprecedented level of human coordination through the help of "the wisdom of the crowd". To work properly, these markets must rely on one or multiple oracles for off-chain event settlement.

2.5.2 Ensuring Oracles honesty As mentioned earlier, blockchains were intentionally built detached from the outer world and its trusted third parties. However, most real-world events still occur outside blockchains, so we need a bridge. While a single corruptible source of truth is unlikely to be sufficient, involving multiple sources of truth is more expensive. So, there is no *one-size-fits-all* solution for this problem and applications need to make the decision of the number of sources they are going to need based on their own need and budgets.

For example, the temperature in Santa Cruz can be reported through a whether forecasting website by a single oracle if it is for a smart contact between our example of Alice and Bob where they have bet 1 BTC each on the outcome of the prediction. In this case a rough precision is tolerated. Nonetheless, if the reported temperature is used to settle a $20M insurance contract, we probably are not going to rely on one or two sources, instead we would involve several, or perhaps many to be precise, oracles such as satellite data, local sensor data, and so on.

Therefore, it is always the matter of trade-offs between cost and decentralisation depending on the value at stake in the solutions offered by oracles.

2.5.3 Centralized solution We already know that it is not easy to guarantee the integrity of a third party. There has been a number of solutions to this well-known problem in the centralised world. The most famous one is probably is the Swiss Cheese Model (see Fig.2). In this model, we use multiple protective layers such as contracts, liable companies, insurances, the law and so on. As long as at least one of these layer works properly, the possible attacks will fail. However, if all layers are penetrated by a attacker, the attack would indeed be successful (Swiss cheese model) (Fig.2).

The takeaway here, as you might have already noticed, is that by using a single corruptable source of truth we are putting the entire system at hazard. Let's see what is the solution for the decentralized world.

2.5.4 Decentralized solution For decentralized world, where we do not have the advantage of having a trusted third part, we can apply the same concept of multiple barriers we discussed for the centralized world to decentralised

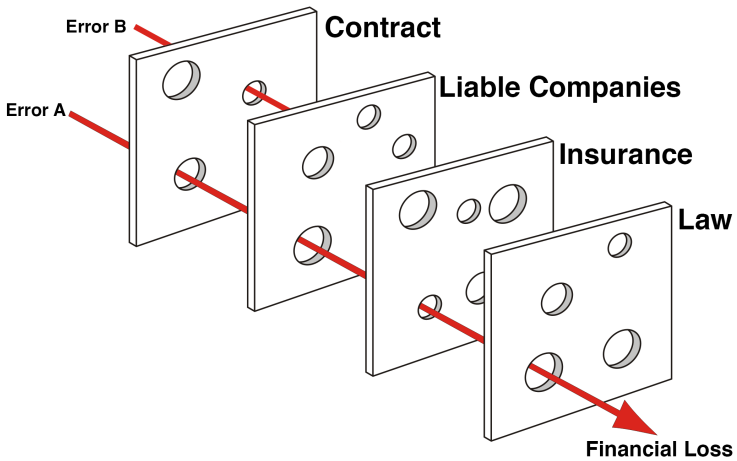

Fig. 2. The Swiss cheese model of accident causation illustrates that, although many layers of defense lie between hazards and accidents, there are flaws in each layer that, if aligned, can allow the accident to occur. [9]

oracles as trust-minimising mechanisms. Here are two main solutions for handling the unreliability of sources in a decentralized world.

A. Multiple data sources This solution might be the most rational and simplest way to decrease the likelihood of getting erroneous data from the outer sources by an oracle. Through this solution, the only way that this strategy might be compromised is either the majority of data sources are compromised, or the oracle itself is compromised (leaving a single point-of-failure).

B. Multiple oracles This solution might be a little bit stronger than the previous one since we increase the number of oracles, each of which uses a number of different sources. They are probabilistically less likely to all be malicious, and therefore the system is safe as long as a majority is honest. Nevertheless, even in this strategy, we are not 100 percent sure about the possible collusion that might happen among the oracles or their sources are compromised.

3 Results and Analysis

3.1 Peer Prediction Markets as Global Games

We have given much more background and explanation on the peering prediction markets. Before comparing it with global games, I would like show an application of global games in economic market which can help to understand better to global game.

Now, we have a binary action global game. Two players are deciding whether to invest. The payoff function is decided by another unknown parameter θ. We show the so many choices as the following table:

Payoffs based on Diff-choices

	Invest	Not Invest
Invest	θ, θ	θ - 1, 0
Not Invest	0, θ - 1	0, 0

Basically, we know that the payoff is depend on the value of θ. When the both players can get the best payoff or some degree of equilibrium. If $\theta < 0$, two players are all have no incentives to invest. Therefore, they will choose not to invest. However, to the case of $\theta > 1$, it is obvious that they'd better to choose both invest. Otherwise, they can get the Nash equilibrium at both invest or both not invest.

The unknown information θ make this game hard to predict. Hence, player i observers their own signal $x_i = \theta + \varepsilon_i$, Like we assumed before, ε_i is normal distribution with the mean of 0. The fluctuation with respect to the mean is known as noise. For now, you may still don't know how to choose an action based on player's private observation. We give a strategy that specify the action with respect to possible private signal.

$$s(x) = \begin{cases} Invest & (x > k) \\ NotInvest & (x \leq k) \end{cases}$$

It is obvious that we switch the strategy around the cutoff value k. It is proved that the cutoff value 1/2 is unique optimal strategy, if we include choose our action which based on other player's action [6]. Hence, the unique equilibrium has both players investing only if they observe a signal which is greater than 1/2.

This is example binary action global game. You may have strong understanding that how global game reached its equilibrium with respect to different conditions. This can also be extended to multiple actions global games. Morries also gave many real applications that model economic problems using global games in [6]. The incomplete information here is some degree similar with peer prediction market which without ground truth.

The underlying contracts of any peer prediction market are ultimately grounded in the behavior of the market itself (or closely related markets). Like the global games not only determined by other players' actions and also the incomplete information which game are they in. Peer prediction implies the contract closing mechanism is decentralized which like the game between different players. Being a market implies that transactions are public and incentives are zero-sum and entirely captured by the market.

3.2 Evaluating Augur for the Peer Prediction Setting

In this section, we are trying to analyze the potential of Augur, which is a decentralized prediction market platform that runs on Ethereum, in peer prediction market.

As we have seen so far, Augur has been almost successful in prediction market through its unique way to handle the possible challenges. However, there has been some ambiguity on how it tries to keep reporters on reporting the truth and not report dishonestly. The question here is that what if the value of the REP that they are going to lose is nothing compared to the value they might earn by reporting wrong. In this way, if the malicious player can somehow succeed in colluding to report the outcome of an even in their own way, the system might fail. This scenario becomes even worse if we apply this method on events whose outcomes are not objectively knowable such as peer prediction market. In other words, only events that have objectively knowable outcomes are suitable for use in Augur markets. In Augur, the reporters report a market as invalid if they believe that the market is not suitable for resolution by the platform. The markets whose outcomes are ambiguous, subjective, or not known outcomes by the event end date can be considered as examples of such invalid markets.

Another scenario that might question the ability of Augur to handle the special cases of real-world markets is when there might be two different legitimate reports on the outcome of an event. For example, in 2006, Trade- Sports created a smart contract to speculate on whether North Korea would fire a ballistic missile that would land outside of its airspace before the end of July 2006. On July 5, 2006, North Korea successfully fired a ballistic missile that landed outside of its airspace, and the event was widely reported by the world media and confirmed by many U.S. government sources. However, the U.S. Department of Defense had not confirmed the event, as was required by TradeSports' contract. TradeSports concluded that the contract's conditions had not been met, and paid out accordingly [7]. As you can clearly see here, in the markets that have a degree of ambiguity such as peer prediction market, Augur might not be able to handle the situation.

3.3 Peer prediction market extension of BTS

In this section, we consider possible extensions of Bayesian Truth Serum using a prediction market. As of writing, we cannot propose a mechanism for closing peer prediction markets with confidence of incentive-compatibility, but we will relate important observations and our ideas.

The critical innovation of the Bayesian Truth Serum is determining which private beliefs are "surprisingly popular" and thus represent private information. In order to make this determination however, BTS requires that participants'

reports to the mechanism designer remain private (to avoid revealing the correct answer to either question to the public).

In considering the marriage of prediction markets with Bayesian Truth Serum, the public operation of a market appears at first glance to be inimicable to the need to maintain this sort of private information aggregation (especially on a public blockchain). But this is not entirely the case, depending on how the market is implemented. We have a few options which fit within the scope of the objectives of this work:

i) Exploit Private Information in Normal Market Operation

Certain information may commonly be considered private in the normal operation of markets and thus exploitable for closing a prediction market. For example:

1. the open market positions of any individual
2. attributes of buyers and sellers on either side of a trade
3. open stop orders

Unfortunately, it is not clear what if any information available is sufficient to recreate a posterior distribution of private beliefs for use with BTS.

ii) Market Simulation We may attempt to elicit enough private information from participants so as to simulate a market (e.g. elicit demand for contracts as a function of market price and calculate equilibrium market price and payouts to participants without any actual trading).

In practice, this looks like generalizing Bayesian Truth Serum to the continuous signals setting and replacing the scoring function used for the *information score* with the payoffs of a prediction market.

3.3.1 Continuous Bayesian Truth Serum with a Simulated Market Consider two separate populations, $\mathcal{G}_1$ and $\mathcal{G}_2$, and a hypothetical Boolean event e which is unobservable, but about which we wish to elicit private beliefs from the populations.

The probability which any agent i assigns to $e = 1$ is p_i. We may elicit these probabilities indirectly from $\mathcal{G}_1$ as described in "Simulating the Market" below. From these values, we calculate

$$q^* = \frac{1}{|\mathcal{G}_1|} \sum_{i \in \mathcal{G}_1} p_i$$

Assume that each participant also assigns a probability density function $q(p)$ to the range of beliefs $p_{j \neq i}$ held by other participants. We elicit from each agent in $\mathcal{G}_2$ the value

$$q_i = \int_0^1 pq(p)\mathrm{d}p$$

As in the standard Bayesian Truth Serum, we calculate the geometric mean of these values as

$$\log \bar{q} = \frac{1}{|\mathcal{G}_2|} \sum_{i \in \mathcal{G}_2} \Big(\log q_i \Big)$$

$\mathcal{G}_2$ will be payed according to a proper scoring rule $S(q_i, q^*)$, while $\mathcal{G}_1$ will be payed according to a simulated prediction market, where the event e is defined according to

$$e := 1(q^* > \bar{q})$$

Simulating the Market

G_1 will be asked to imagine participating in a standard prediction market for a single type of contract which offer a payout of $\$1$ conditional on event e. For a number of hypothetical market prices $0 \leq \pi \leq 1$, we ask for the number of contracts v_i each agent would be willing to buy or short sell.

Assuming a logarithmic utility function for all participants, it may be shown that expected utility is maximized by

$$w_i \frac{p_i - \pi}{(1-\pi)\pi} = v_i$$

where w_i is the wealth of agent i and π is the available price of a trade. By eliciting numerous (p_i, v_i) samples, we may fit for an agent's private belief p_i and wealth w_i.

Under the logarithmic utility assumption, the equilibrium price of the market will be the wealth-weighted average of private beliefs regarding e:

$$\pi^* = \frac{\sum_{i \in \mathcal{G}_1} w_i p_i}{\sum_{i \in \mathcal{G}_1} w_i}$$

Which can be calculated by our fitted values for w_i and p_i. Closing the hypothetical market at π^* yields payoff for each agent $i \in \mathcal{G}_1$ of

$$w_i \frac{p_i - \pi^*}{(1 - \pi^*)\pi^*} \left(e - \pi^* \right)$$

which may be negative.

Though we incentive-compatibility may be a feature of this proposed mechanism, we have not proven it.

4 Conclusion

In this work, we have first considered peer prediction markets as instances of global games. Prediction market can be expressed by global games on certain conditions, such as when the information used to close contracts is private. Moreover, we have evaluated Augur's mechanisms for use with a general peer prediction market. As a result, Augur are still not mature enough to handle the rising problems on peer prediction market. We have proposed possible extensions of Bayesian Truth Serum which may be more appropriate for "peer-prediction markets", albeit without proof of incentive compatibility and still requiring a central mechanism designer to pay participants, but which which may be fruitful for future investigation.

5 Team member Contributions

Please describe the contribution of each member in you group.

- **Saeed:** Presented hash functions and public key cryptography for in-class presentation on Bitcoin. Summarized Augur and evaluated for peer prediction setting for final presentation and in this report. Initiated preparation of slides for in-class presentation on Bitcoin and template of this final report.
- **Reilly:** Presented blockchain concepts and proof-of-work for in-class presentation on Bitcoin. Wrote proposal for final project and organized team. Provided background and results for market implementation of BTS in class and in this final report.
- **Yanwen:** Took on responsibility for introduction and conclusion for each of our presentations (Bitcoin and final presentation) and this final report. Initiated preparation of slides for final in-class presentation of this project.
- **Jie:** Presented incentives and attacks on blockchain for our in-class presentation on Bitcoin, and linked peer prediction markets to global games for final presentation and this report.

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